Ear and Labyrinth Disorders	
Uncommon	Ear pain, Tinnitus
Respiratory, Thoracic and Mediastinal Disorders	
Common	Dyspnea, Epistaxis, Cough, Nasal congestion, Pharyngolaryngeal pain
Uncommon	Wheezing, Pneumonia aspiration, Pulmonary congestion, Respiratory disorder, Rales, Respiratory tract congestion, Dysphonia
Rare	Sleep apnea syndrome, Hyperventilation
Gastrointestinal Disorders	
Common	Vomiting, Diarrhea, Constipation, Nausea, Abdominal pain, Dyspepsia, Dry mouth, Stomach discomfort
Uncommon	Dysphagia, Gastritis, Fecal incontinence, Fecaloma
Rare	Intestinal obstruction, Pancreatitis, Lip swelling, Cheilitis
Renal and Urinary Disorders	
Common	Enuresis
Uncommon	Urinary retention, Dysuria, Urinary incontinence, Pollakiuria
Skin and Subcutaneous Tissue Disorders	
Common	Rash, Erythema
Uncommon	Angioedema, Skin lesion, Skin disorder, Pruritus, Acne, Skin discolouration, Alopecia, Seborrhoeic dermatitis, Dry skin, Hyperkeratosis
Rare	Dandruff
Musculoskeletal and Connective Tissue Disorders	
Common	Arthralgia, Back pain, Pain in extremity
Uncommon	Muscular weakness, Myalgia, Neck pain, Joint swelling, Posture abnormal, Joint stiffness, Musculoskeletal chest pain
Rare	Rhabdomyolysis
Endocrine Disorders	
Rare	Inappropriate antidiuretic hormone secretion
Metabolism and Nutrition Disorders	
Common	Increased appetite, Decreased appetite
Uncommon	Diabetes mellitus, Anorexia, Polydipsia, Hyperglycemia, Blood cholesterol increased, Blood triglycerides increased
Rare	Hypoglycemia
Very rare	Diabetic ketoacidosis
Not known	Water intoxication
Infections and Infestations	
Common	Pneumonia, Influenza, Bronchitis, Upper respiratory tract infection, Urinary tract infection
Uncommon	Sinusitis, Viral infection, Ear infection, Tonsillitis, Cellulitis, Otitis media, Eye infection, Localised infection, Acarodermatitis, Respiratory tract infection, Cystitis, Onychomycosis
Rare	Otitis media chronic
Vascular Disorders	
Uncommon	Hypotension, Orthostatic hypotension, Flushing
General Disorders and Administration Site Conditions	
Common	Pyrexia, Fatigue, Peripheral edema, Asthenia, Chest pain
Uncommon	Face edema, Gait disturbance, Feeling abnormal, Sluggishness, Influenza like illness, Thirst, Chest discomfort, Chills
Rare	Generalised edema, Hypothermia, Drug withdrawal syndrome, Peripheral coldness
Immune System Disorders	
Uncommon	Hypersensitivity
Rare	Drug hypersensitivity
Not known	Anaphylactic reaction
Hepatobiliary Disorders	
Rare	Jaundice
Pregnancy, Puerperium and Perinatal Conditions	
Not known	Drug withdrawal syndrome neonatal
Reproductive System and Breast Disorders	
Uncommon	Amenorrhea, Sexual dysfunction, Erectile dysfunction, Ejaculation disorder, Galactorrhea, Gynecomastia, Menstrual disorder, Vaginal discharge,
Not known	Priapism
Psychiatric Disorders	
Very common	Insomnia
Common	Anxiety, Agitation, Sleep disorder
Uncommon	Confusional state, Mania, Libido decreased, Listless, Nervousness
Rare	Anorgasmia, Blunted affect

Class Effects

As with other antipsychotics, very rare cases of QT prolongation have been reported with risperidone. Other class-related cardiac effects reported with antipsychotics which prolong QT interval include ventricular arrhythmia, ventricular fibrillation, ventricular tachycardia, sudden death, cardiac arrest and Torsades de Pointes.

Other additional reported adverse reactions are rhinorrhea, rhinitis, abdominal discomfort, nasopharyngitis, eyelid edema, eyelid margin crusting, aptyalism, malaise, pitting edema, edema, pharyngitis, bronchopneumonia, tracheobronchitis, hematocrit decreased, dysarthria, middle insomnia, menstruation irregular, breast enlargement, sinus congestion, nasal edema, rash erythematous, rash papular, rash generalized, rash maculopapular, dysgeusia, menstruation delayed, ejaculation delayed, oligomenorrhea, breast discomfort, glucose present in urine, organic dysfunction, pituitary adenoma, pulmonary embolism, precocious puberty, cardiopulmonary arrest, and sudden death, thrombotic thrombocytopenic purpura, weight decreased, gamma-glutamyltransferase increased, hepatic enzyme increased, bradycardia, neutropenia, paresthesia, convulsion, blepharospasm, vertigo, toothache, tongue spasm, eczema, buttock pain, lower respiratory tract infection, infection, gastroenteritis, subcutaneous abscess, fall, hypertension, pain, depression.

OVERDOSE

Symptoms

In general, reported signs and symptoms have been those resulting from an exaggeration of risperidone known pharmacological effects. These include drowsiness and sedation, tachycardia and hypotension, and extrapyramidal symptoms. In overdose, QT-prolongation and convulsions have been reported. Torsade de pointes has been reported in association with combined overdose of risperidone and paroxetine.

In case of acute overdosage, the possibility of multiple drug involvement should be considered.

Treatment

Establish and maintain a clear airway, and ensure adequate oxygenation and ventilation. Gastric lavage (after intubation, if the patient is unconscious) and administration of activated charcoal together with a laxative should be considered. Cardiovascular monitoring should commence immediately and should include continuous electrocardiographic monitoring to detect possible arrhythmias.

There is no specific antidote to risperidone. Therefore, appropriate supportive measures should be instituted. Hypotension and circulatory collapse should be treated with appropriate measures such as intravenous fluids and/or sympathomimetic agents. In case of severe extrapyramidal symptoms, anticholinergic medication should be administered. Close medical supervision and monitoring should continue until the patient recovers.

STORAGE AND HANDLING INSTRUCTIONS

Store below 30°C, protected from light and moisture.

PACKAGING

ROZIDAL TABLETS

Blister pack of 10's; Box of 10 x 10's, Box of 6 x 10's
Blister strip of 10's; Box of 10 x 10's, Box of 6 x 10's

ZESPERONE MD TABLETS

Blister strip of 10's; Box of 10 x 10's, Box of 6 x 10's

Revised date: October 2017

Product Registration Holder / Manufactured by:
RANBAXY (MALAYSIA) SDN. BHD.
Lot 23, Bakar Arang Industrial Estate
08000 Sungai Petani, Kedah.
Malaysia.

5189057

For the use only of a Psychiatrist
PRESCRIBING INFORMATION

ROZIDAL TABLETS / ZESPERONE MD TABLETS

(Risperidone Tablets 1/2/3 mg & Risperidone Mouth Dissolving Tablets 1/2mg)

> WARNING: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA RELATED PSYCHOSIS
> Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone is not approved for the treatment of patients with dementia-related psychosis. (see WARNINGS AND PRECAUTIONS)

COMPOSITION

ROZIDAL TABLETS 1 mg
Each tablet contains
Risperidone ....1 mg

ROZIDAL TABLETS 2 mg
Each tablet contains
Risperidone ....2 mg

ROZIDAL TABLETS 3 mg
Each tablet contains
Risperidone ....3 mg

Excipients: Lactose Monohydrate, Pregelatinised Starch, Microcrystalline Cellulose, Sodium Lauryl Sulphate, Croscarmellose Sodium, Colloidal Silicon Dioxide, Magnesium Stearate, Opadry White, Opadry Orange, Opadry Yellow, Purified Water

ZESPERONE MD TABLETS 1 mg
Each orodispersible tablet contains:
Risperidone ....1 mg

ZESPERONE MD TABLETS 2 mg
Each orodispersible tablet contains:
Risperidone ....2 mg

Excipients: Magnesium Stearate, Ferric Oxide (Red), Mannitol, Saccharin sodium, Heavy magnesium carbonate, Flavour Peppermint 517 (spray dried), Colloidal Anhydrous Silica, Levomenthol, Aspartame, Croscarmellose Sodium, Low substituted hydroxypropyl cellulose, Purified Talc.

PRODUCT DESCRIPTION

ROZIDAL TABLETS 1 mg
White capsule shaped, biconvex film coated tablets debossed with 'RA85' on one side and plain on the other side.

ROZIDAL TABLETS 2 mg
Orange coloured, capsule shaped, biconvex film coated tablets debossed with 'RA86' on one side and plain on the other side.

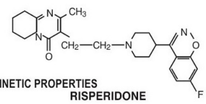
ROZIDAL TABLETS 3 mg
Yellow coloured, capsule shaped, biconvex film coated tablets debossed with 'RA87' on one side and plain on the other side.

ZESPERONE MD TABLETS 1 mg
Pink coloured, peppermint flavoured, mottled, circular, flat beveled tablets, debossed with 'R' on one side and '1' on the other side.

ZESPERONE MD TABLETS 2 mg
Pink coloured, peppermint flavoured, mottled, circular, flat beveled tablets, debossed with 'R' on one side and '2' on the other side.

DESCRIPTION

ROZIDAL TABLETS/ ZESPERONE MD TABLETS contain risperidone. Risperidone is a psychotropic agent belonging to the chemical class of benzisoxazole derivatives. It is chemically designated as, 3-[2-[4-(6-fluoro-1,2-benzisoxazol-3-yl)-1-piperidinyl]ethyl]-6,7,8,9 tetrahydro- 2-methyl-4H-pyrido[1,2-a]pyrimidin-4-one. The empirical formula of risperidone is $C_{23}H_{27}FN_4O_2$ and its molecular weight is 410.49. Its structural formula is as follows



RISPERIDONE

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

• Mechanism of Action

Risperidone is a selective monoaminergic antagonist with unique properties. It has a high affinity for serotoninergic 5-HT_2 and dopaminergic D_2-receptors. Risperidone binds also to α_1-adrenergic receptors, and, with lower affinity, to H_1-histaminergic and α_2-adrenergic receptors. Risperidone has no affinity for cholinergic receptors. Although risperidone is a potent D_2-antagonist, which is considered to improve the positive symptoms of schizophrenia, it causes less depression of motor activity and induction of catalepsy than classical neuroleptics. Balanced central serotonin and dopamine antagonism may reduce extrapyramidal side effect liability and extend the therapeutic activity to the negative and affective symptoms of schizophrenia.

• Pharmacokinetics

Risperidone is completely absorbed after oral administration, reaching peak plasma concentrations within 1-2 hrs. The absorption is not affected by food and thus, it can be given with or without meals.

Steady state of risperidone is reached within 1 day in most patients. Steady state of 9-hydroxy-risperidone is reached within 4-5 days of dosing. Risperidone plasma concentrations are dose-proportional within the therapeutic dose range.

Risperidone is rapidly distributed. The volume of distribution is 1-2 L/kg. In the plasma, risperidone is bound to albumin and α_1-acid glycoprotein. The plasma protein-binding of risperidone is 88%; that of 9-hydroxy-risperidone is 77%.

Risperidone is metabolised by CYP 2D6 to 9-hydroxy-risperidone, which has a similar pharmacological activity as risperidone. Risperidone plus 9-hydroxy-risperidone form the active antipsychotic fraction. CYP 2D6 is subject to genetic polymorphism. Extensive CYP 2D6 metabolisers convert risperidone rapidly into 9-hydroxy-risperidone, whereas poor CYP 2D6 metabolisers convert it much more slowly. Although extensive metabolisers have lower risperidone and higher 9-hydroxy-risperidone concentrations than poor metabolisers, the pharmacokinetics of risperidone and 9-hydroxy-risperidone combined (i.e., the active antipsychotic fraction), after single and multiple doses, are similar in extensive and poor metabolisers of CYP 2D6.

Another metabolic pathway of risperidone is N-dealkylation. Risperidone at clinically relevant concentration does not substantially inhibit the metabolism of medicines metabolised by cytochrome P450 isozymes, including CYP 1A2, CYP 2A6, CYP 2C8/9/10, CYP 2D6, CYP 2E1, CYP 3A4, and CYP 3A5 *in vitro*. One week after administration, 70% of the dose is excreted in the urine and 14% in the faeces. In the urine, risperidone plus 9-hydroxy-risperidone represent 35-45% of the dose. The remainders are inactive metabolites. After oral administration to psychotic patients, risperidone is eliminated with a half-life of about 3 hours. The elimination half-life of 9-hydroxy-risperidone and of the active antipsychotic fraction is 24 hours.

Higher active plasma concentrations and a reduced clearance of the active antipsychotic fraction by 30% in the elderly and 60% in patients with renal insufficiency have been reported after single dose. Risperidone plasma concentrations were normal in patients with liver insufficiency, but the mean free fraction of risperidone in plasma was increased by about 35%.

The pharmacokinetics of risperidone, 9-hydroxy-risperidone and the active antipsychotic fraction in children are similar to those in adults.

Gender, race and smoking habits

A population pharmacokinetic analysis reported no apparent effect of gender, race or smoking habits on the pharmacokinetics of risperidone or the active antipsychotic fraction.

INDICATIONS

ROZIDAL TABLETS/ ZESPERONE MD TABLETS are indicated for the treatment of a broad range of patients with schizophrenia, including first episode psychoses, acute schizophrenic exacerbations, chronic schizophrenia, and other psychotic conditions, in which positive symptoms (such as hallucinations, delusions, thought disturbances, hostility, suspiciousness) and/or negative symptoms (such as blunted affect, emotional and social withdrawal, speech difficulty) are prominent. Risperidone alleviates affective symptoms (such as depression, guilt feelings, anxiety) associated with schizophrenia.

Risperidone is also effective in maintaining the clinical improvement during continuation therapy in patients who have shown an initial treatment response.

DOSE AND METHOD OF ADMINISTRATION

ROZIDAL TABLETS are available as 1 mg, 2 mg & 3 mg strengths and **ZESPERONE MD TABLETS** are available as 1 mg and 2 mg strengths which may not be suitable for all dosage recommendations given below. In such cases, other suitable approved dosage forms of risperidone (such as oral solution) or suitable strengths (such as 0.5 mg film-coated or mouth dissolving tablets) should be used.

Schizophrenia

Switching From Other Antipsychotics: When medically appropriate, gradual discontinuation of the previous treatment while risperidone therapy is initiated is recommended. Also if medically appropriate, when switching patients from depot antipsychotics, initiate risperidone therapy in place of the next scheduled injection. The need for continuing existing anti-parkinson medications should be re-evaluated periodically.

Adults: Risperidone may be given once daily or twice daily.

Patients should start with 2 mg/day risperidone. The dosage may be increased on the second day to 4 mg. From then on the dosage can be maintained unchanged or further individualized, if needed. Most patients will benefit from daily doses between 4 and 6 mg. In some patients, a slower titration phase and a lower starting and maintenance dose may be appropriate.

Doses above 10 mg/day have not been shown to be superior in efficacy to lower doses and may cause extrapyramidal symptoms. Since the safety of doses above 16 mg/day has not been evaluated, doses above this level should not be used.

A benzodiazepine may be added to risperidone when additional sedation is required.

***Method of Administration* for ZESPERONE MD TABLETS**

Do not open the blister until ready to administer. Remove the tablet from the blister with dry hands.
Immediately place the tablet on the tongue. The tablet will begin disintegrating within seconds. Water may be used if desired.

Use in Special Population

• Pediatrics

Experience in schizophrenia is lacking in children <15 years.

• Geriatrics

A starting dose of 0.5 mg twice daily is recommended. This dosage can be individually adjusted with 0.5 mg twice daily increments to 1-2 mg twice daily.

Risperidone is well tolerated by the elderly.

• Renal and Hepatic Impairment

A starting dose of 0.5 mg twice daily is recommended. This dosage can be individually adjusted with 0.5 mg twice daily increments to 1-2 mg twice daily.

Risperidone should be used with caution in this group of patients until further experience is gained.

USE IN SPECIAL POPULATION

• Pregnancy and Lactation

Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalization.

There was one report of a case of agenesis of the corpus callosum in an infant exposed to risperidone *in utero*. The causal relationship to risperidone therapy is unknown.

Risperidone should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

The safety of risperidone for use during human pregnancy has not been established. In experimental animals, risperidone did not show direct reproductive toxicity, some indirect, prolactin- and CNS-mediated effects were observed. No teratogenic effect of risperidone was reported.

The effect of risperidone on labor and delivery in humans is unknown.

In animals, risperidone and 9-hydroxyrisperidone have been reported to be excreted in the milk. It has been reported that risperidone and 9-hydroxyrisperidone are also excreted in human breast milk. Therefore, women receiving risperidone should not breast feed.

CONTRAINDICATIONS

ROZIDAL TABLETS/ ZESPERONE MD TABLETS are contraindicated in patients with a known hypersensitivity to risperidone or any of the components of the formulation.

It is also contraindicated in coma caused by CNS depressants, bone marrow depression and avoided in phaeochromocytoma.

WARNINGS AND PRECAUTIONS

Intraoperative Floppy Iris Syndrome

Intraoperative floppy iris syndrome (IFIS) has been observed during cataract surgery in patients treated with medicines with alpha1a-adrenergic antagonist effect, including risperidone. IFIS may increase the risk of eye complications during and after the operation. Current or past use of medicines with alpha1a-adrenergic antagonist effect should be made known to the ophthalmic surgeon in advance of surgery. The potential benefit of stopping alpha1 blocking therapy prior to cataract surgery has not been established and must be weighed against the risk of stopping the antipsychotic therapy.

Hyperglycemia and Diabetes Mellitus

Hyperglycemia in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients treated with atypical antipsychotics. Assessment of the relationship between atypical antipsychotic use and glucose abnormalities is complicated by the possibility of an increased background risk of diabetes mellitus in patients with schizophrenia and the increasing incidence of diabetes mellitus in the general population. Given this confounders, the relationship between atypical antipsychotic use and hyperglycemia-related adverse events is not completely understood. However, epidemiological data suggest an increased risk of treatment-emergent hyperglycemia-related events in patients treated with the atypical antipsychotics. Precise risk estimates for hyperglycemia-related adverse events in patients treated with atypical antipsychotics are not available.

Patients with an established diagnosis of diabetes mellitus who are started on atypical antipsychotics should be monitored regularly for worsening of glucose control. Patients with risk factors for diabetes mellitus (e.g. obesity, family history of diabetes) who are starting treatment with atypical antipsychotics should undergo fasting blood glucose testing at the beginning of treatment and periodically during treatment. Any patient treated with atypical antipsychotics should be monitored for symptoms of hyperglycemia including polydipsia, polyuria, polyphagia, and weakness. Patients who develop symptoms of hyperglycemia during treatment with atypical antipsychotics should undergo fasting blood glucose testing. In some cases, hyperglycemia has resolved when the atypical antipsychotic was discontinued; however, some patients required continuation of anti-diabetic treatment despite discontinuation of the suspect drug.

Elderly Patients with Dementia

Increased mortality has been reported in elderly patients with dementia treated with atypical antipsychotic drugs, including risperidone. The incidence of mortality has been reported to be 4.0% for risperidone-treated patients. The mean age (range) of patients who died was 86 years (range 67 - 100).

Concomitant Use with Furosemide

A higher incidence of mortality has been reported in patients treated with furosemide plus risperidone when compared to patients treated with risperidone alone or furosemide alone.

No pathophysiological mechanism has been identified to explain this finding, and no consistent pattern for cause of death observed. Nevertheless caution should be exercised and the risks and benefits of this combination should be considered prior to the decision to use. There was no increased incidence of mortality among patients taking other diuretics as concomitant medication with risperidone. Irrespective of treatment, dehydration was an overall risk factor for mortality and should therefore be carefully avoided in elderly patients with dementia.

Cerebrovascular Adverse Events (CVAE)

In elderly patients with dementia, a higher incidence of cerebrovascular adverse events, (cerebrovascular accidents and transient ischemic attacks), including fatalities, has been reported in patients treated with risperidone.

The mechanism for this increased risk is not known. An increased risk cannot be excluded for other antipsychotics or other patient populations. Risperidone should be used with caution in patients with risk factors for stroke.

The risk of CVAEs was significantly higher in patients with mixed or vascular type of dementia when compared to Alzheimer's dementia. Therefore, patients with other types of dementias than Alzheimer's should not be treated with risperidone.

Physicians are advised to assess the risks and benefits of the use of risperidone in elderly patients with dementia, taking into account risk predictors for stroke in the individual patient. Patients/caregivers should be cautioned to immediately report signs and symptoms of potential CVAEs such as sudden weakness or numbness in the face, arms or legs, and speech or vision problems. All treatment options should be considered without delay, including discontinuation of risperidone.

Risperidone should only be used short term for persistent aggression in patients with moderate to severe Alzheimer's dementia to supplement non-pharmacological approaches which have had limited or no efficacy and when there is potential risk of harm to self or others.

Patients should be reassessed regularly, and the need for continuing treatment reassessed.

Alpha-blocking Activity

Due to the alpha-blocking activity of risperidone, orthostatic hypotension can occur, especially during the initial dose-titration period. Clinically significant hypotension has been observed post-marketing with concomitant use of risperidone and antihypertensive treatment. Risperidone should be used with caution in patients with known cardiovascular disease (e.g., heart failure, myocardial infarction, conduction abnormalities, dehydration, hypovolemia, or cerebrovascular disease), and the dosage should be gradually titrated as recommended. A dose reduction should be considered if hypotension occurs.

Tardive Dyskinesia/Extrapyramidal Symptoms (TD/EPS)

Drugs with dopamine receptor antagonistic properties have been associated with the induction of tardive dyskinesia, characterised by rhythmical involuntary movements, predominantly of the tongue and/or face. It has been reported that the occurrence of extrapyramidal symptoms is a risk factor for the development of tardive dyskinesia. Because risperidone has a lower potential to induce extrapyramidal symptoms than classical neuroleptics, it should have a reduced risk of inducing tardive dyskinesia as compared to classical neuroleptics. If signs and symptoms of tardive dyskinesia appear, the discontinuation of all antipsychotic drugs should be considered.

Neuroleptic Malignant Syndrome (NMS)

Neuroleptic malignant syndrome, characterised by hyperthermia, muscle rigidity, autonomic instability, altered consciousness and elevated CPK levels, has been reported to occur with antipsychotics. Additional signs may include myoglobinuria (rhabdomyolysis) and acute renal failure. In this event all antipsychotic drugs including risperidone should be discontinued.

The diagnostic evaluation of patients with this syndrome is complicated. In arriving at a diagnosis, it is important to identify cases in which the clinical presentation includes both serious medical illness (e.g., pneumonia, systemic infection, etc.) and untreated or inadequately treated extrapyramidal signs and symptoms (EPS). Other important considerations in the differential diagnosis include central anticholinergic toxicity, heat stroke, drug fever, and primary central nervous system pathology.

The management of NMS should include: (1) immediate discontinuation of antipsychotic drugs and other drugs not essential to concurrent therapy; (2) intensive symptomatic treatment and medical monitoring; and (3) treatment of any concomitant serious medical problems for which specific treatments are available. There is no general agreement about specific pharmacological treatment regimens for uncomplicated NMS.

If a patient requires antipsychotic drug treatment after recovery from NMS, the potential reintroduction of drug therapy should be carefully considered. The patient should be carefully monitored, since recurrences of NMS have been reported.

Physicians should weigh the risks versus the benefits when prescribing antipsychotics, including risperidone, to patients with Parkinson's Disease or Dementia with Lewy

Bodies (DLB) since both groups may be at increased risk of Neuroleptic Malignant Syndrome as well as having an increased sensitivity to antipsychotic medications. Manifestation of this increased sensitivity can include confusion, obtundation, postural instability with frequent falls, in addition to extrapyramidal symptoms.

Weight Gain

Significant weight gain has been reported. Monitoring weight gain is advisable when risperidone is being used. Patients may be advised to refrain from excessive eating in view of the possibility of weight gain.

Hyperprolactinaemia

Risperidone should be used with caution in patients with pre-existing hyperprolactinaemia and in patients with possible prolactin-dependent tumours.

Dyslipidemia

Undesirable alterations in lipids have been reported in patients treated with atypical antipsychotics.

QT Prolongation

QT prolongation has very rarely been reported postmarketing. As with other antipsychotics, caution should be exercised when risperidone is prescribed in patients with known cardiovascular disease, family history of QT prolongation, bradycardia, or electrolyte disturbances (hypokalaemia, hypomagnesaemia), as it may increase the risk of arrhythmogenic effects, and in concomitant use with medicines known to prolong the QT interval.

Priapism

Priapism may occur with risperidone treatment due to its alpha-adrenergic blocking effects.

Body Temperature Regulation

Disruption of the body's ability to reduce core body temperature has been attributed to antipsychotic medicines. Appropriate care is advised when prescribing risperidone to patients who will be experiencing conditions which may contribute to an elevation in core body temperature, e.g., exercising strenuously, exposure to extreme heat, receiving concomitant treatment with anticholinergic activity, or being subject to dehydration.

Venous Thromboembolism

Cases of venous thromboembolism (VTE) have been reported with antipsychotic drugs. Since patients treated with antipsychotics often present with acquired risk factors for VTE, all possible risk factors for VTE should be identified before and during treatment with risperidone and preventative measures undertaken.

Leukopenia, Neutropenia, and Agranulocytosis

Class Effect: Events of leukopenia/neutropenia have been reported temporally related to antipsychotic agents, including risperidone. Agranulocytosis has also been reported.

Possible risk factors for leukopenia/neutropenia include pre-existing low white blood cell count (WBC) and history of drug-induced leukopenia/neutropenia. Patients with a history of a clinically significant low WBC or a drug-induced leukopenia/neutropenia should have their complete blood count (CBC) monitored frequently during the first few months of therapy and discontinuation of risperidone should be considered at the first sign of a clinically significant decline in WBC in the absence of other causative factors.

Patients with clinically significant neutropenia should be carefully monitored for fever or other symptoms or signs of infection and treated promptly if such symptoms or signs occur. Patients with severe neutropenia (absolute neutrophil count <1000/mm^3) should discontinue risperidone and have their WBC followed until recovery.

Potential for Cognitive and Motor Impairment

Somnolence was a commonly reported adverse event associated with risperidone treatment, especially when ascertained by direct questioning of patients. This adverse event is dose-related, and somnolence has been reported significantly in high-dose patients (risperidone 16 mg/day). Direct questioning is more sensitive for detecting adverse events than spontaneous reporting. Since risperidone has the potential to impair judgment, thinking, or motor skills, patients should be cautioned about operating hazardous machinery, including automobiles, until they are reasonably certain that risperidone therapy does not affect them adversely.

Dysphagia

Esophageal dysmotility and aspiration have been associated with antipsychotic drug use. Aspiration pneumonia is a common cause of morbidity and mortality in patients with advanced Alzheimer's dementia. risperidone and other antipsychotic drugs should be used cautiously in patients at risk for aspiration pneumonia.

Effects on Ability to Drive and Use Machines

Risperidone may interfere with activities requiring mental alertness. Therefore, patients should be advised not to drive or operate machinery until their individual susceptibility is known.

Other

Classical neuroleptics are known to lower the seizure threshold. Caution is recommended when treating patients with epilepsy.

ROZIDAL TABLETS contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

ZESPERONE MD TABLETS contains a source of phenylalanine which may be harmful for people with phenylketonuria.

DRUG INTERACTIONS

Given the primary CNS effects of risperidone, it should be used with caution in combination with other centrally acting drugs and alcohol.

Risperidone may antagonise the effect of levodopa and other dopamine-agonists. If this combination is deemed necessary, particularly in end-stage Parkinson's disease, the lowest effective dose of each treatment should be prescribed.

Clinically significant hypotension has been observed postmarketing with concomitant use of risperidone and antihypertensive treatment.

As with other antipsychotics, caution is advised when prescribing risperidone with medicinal products known to prolong the QT interval, e.g., class Ia antiarrhythmics (e.g., quinidine, dysopiramide, procainamide), class III antiarrhythmics (e.g., amiodarone, sotalol), tricyclic antidepressant (i.e., amitriptyline), tetracyclic antidepressants (i.e., maprotiline), some antihistaminics, other antipsychotics, some antimalarials (i.e., chinice and mefloquine), and with medicines causing electrolyte imbalance (hypokalaemia, hypomagnesiaemia), bradycardia, or those which inhibit the hepatic metabolism of risperidone. This list is indicative and not exhaustive. Caution is advised when prescribing risperidone with drugs known to prolong the QT interval.

Risperidone does not show a clinically relevant effect on the pharmacokinetics of lithium, valproate, digoxin or topiramate. Topiramate modestly reduced the bioavailability of risperidone, but not that of the active antipsychotic fraction. Therefore, this interaction is unlikely to be of clinical significance.

Fluoxetine and paroxetine, CYP2D6 inhibitors, may increase the plasma concentration of risperidone but less so of the active antipsychotic fraction. It is expected that other CYP 2D6 inhibitors, such as quinidine, may affect the plasma concentrations of risperidone in a similar way. When concomitant fluoxetine or paroxetine is initiated or discontinued, the physician should re-evaluate the dosing of risperidone.

Carbamazepine has been shown to decrease the plasma levels of the antipsychotic fraction of risperidone. Similar effects may be observed with e.g. rifampicin, phenytoin and phenobarbital which also induce CYP 3A4 hepatic enzyme as well as P-glycoprotein. When carbamazepine or other CYP 3A4 hepatic enzyme/P-glycoprotein (P-gp) inducers are initiated or discontinued, the physician should re-evaluate the dosing of risperidone.

Phenothiazines, tricyclic antidepressants and some beta-blockers may increase the plasma concentrations of risperidone but not those of the active antipsychotic fraction. Amitriptyline does not affect the pharmacokinetics of risperidone or the active antipsychotic fraction. Cimetidine and ranitidine increase the bioavailability of risperidone, but only marginally that of the active antipsychotic fraction. Erythromycin, a CYP 3A4 inhibitor, does not change the pharmacokinetics of risperidone and the active antipsychotic fraction. The cholinesterase inhibitors, galantamine and donepezil, do not show a clinically relevant effect on the pharmacokinetics of risperidone and the active antipsychotic fraction. When risperidone is taken together with other highly protein-bound drugs, there is no clinically relevant displacement of either drug from the plasma proteins.

Verapamil, an inhibitor of CYP 3A4 and P-gp, increases the plasma concentration of risperidone.

Food does not affect the absorption of risperidone.

Chronic administration of clozapine with risperidone may decrease the clearance of risperidone.

The combined use of psychostimulants (e.g., methylphenidate) with risperidone in children and adolescents did not alter the pharmacokinetics and efficacy of risperidone.

Concomitant use of oral risperidone with paliperidone is not recommended as paliperidone is the active metabolite of risperidone and the combination of the two may lead to additive active antipsychotic fraction exposure.

UNDESIRABLE EFFECTS

Side Effects

Postmarketing Data

Eye Disorders

Frequency: Not known – Floppy iris syndrome (intraoperative)

Based on extensive clinical experience including long term use, risperidone is generally well tolerated. In many instances it has been difficult to differentiate adverse events from symptoms of the underlying disease. Adverse events observed in association with the use risperidone are listed below.

The most frequently reported adverse drug reactions (ADRs) are: Parkinsonism, headache, and insomnia.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Adverse Drug Reactions by System Organ Class and Frequency	
Investigations	
Common	Blood prolactin increased[a], Weight increased
Uncommon	Electrocardiogram QT prolonged, Electrocardiogram abnormal, Transaminases increased, White blood cell count decreased, Body temperature increased, Eosinophil count increased, Hemoglobin decreased, Blood creatine phosphokinase increased
Rare	Body temperature decreased
[a] Hyperprolactinemia can in some cases lead to gynecomastia, menstrual disturbances, amenorrhea, galactorrhea	
Cardiac Disorders	
Common	Tachycardia
Uncommon	Atrioventricular block, Bundle branch block, Atrial fibrillation, Sinus bradycardia, Palpitations
Blood and Lymphatic System Disorders	
Uncommon	Neutropenia, Anemia, Thrombocytopenia
Rare	Granulocytopenia
Not known	Agranulocytosis
Nervous System Disorders	
Very common	Parkinsonism[b], Headache
Common	Akathisia[b], Dizziness, Tremor[b], Dystonia[b], Somnolence, Sedation, Lethargy, Dyskinesia[b]
Uncommon	Unresponsive to stimuli, Loss of consciousness, Syncope, Depressed level of consciousness, Cerebrovascular accident, Transient ischemic attack, Dysarthria, Disturbance in attention, Hypersomnia, Dizziness postural, Balance disorder, Tardive dyskinesia, Speech disorder, Coordination abnormal, Hypoesthesia, Dysgeusia
Rare	Neuroleptic malignant syndrome, Diabetic coma, Cerebrovascular disorder, Cerebral ischemia, Movement disorder, Head titubation
[b] Extrapyramidal disorder may occur: Parkinsonism (salivary hypersecretion, musculoskeletal stiffness, parkinsonism, drooling, cogwheel rigidity, bradykinesia, hypokinesia, masked facies, muscle tightness, akinesia, nuchal rigidity, muscle rigidity, parkinsonian gait, and glabellar reflex abnormal), akathisia (akathisia, restlessness, hyperkinesia, and restless leg syndrome), tremor, dyskinesia (dyskinesia, muscle twitching, choreoathetosis, athetosis, and myoclonus), dystonia. Dystonia includes dystonia, muscle spasms, hypertonia, torticollis, muscle contractions involuntary, muscle contracture, blepharospasm, oculogyration, tongue paralysis, facial spasm, laryngospasm, myotonia, opisthotonus, oropharyngeal spasm, pleurothotonus, tongue spasm, and trismus. Tremor includes tremor and parkinsonian rest tremor. It should be noted that a broader spectrum of symptoms are included, that do not necessarily have an extrapyramidal origin	
Eye Disorders	
Common	Vision blurred
Uncommon	Conjunctivitis, Ocular hyperaemia, Eye discharge, Eye swelling, Dry eye, Lacrimation increased, Photophobia
Rare	Visual acuity reduced, Eye rolling, Glaucoma